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The Influence of Maturity Status on Drop Jump Kinetics in Male Youth

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¹Youth Physical Development Center, Cardiff School of Sport and Health Sciences, Cardiff Metropolitan University, Cardiff, United Kingdom; ²Sport Performance Research Institute, New Zealand (SPRINZ), AUT University, Auckland, New Zealand; and ³Centre for Sport Science and Human Performance, Waikato Institute of Technology, Hamilton, New Zealand

Abstract

Kumar, NTA, Radnor, JM, Oliver, JL, Lloyd, RS, CSCSD, Pedley, JS, Wong, MA, and Dobbs, IJ. The influence of maturity status on drop jump kinetics in male youth. *J Strength Cond Res* 38(1): 38–46, 2024—The aim of this study was to examine the effects of maturity status on drop jump (DJ) kinetics in young male athletes (categorized as early-pre-peak height velocity [PHV] [< -2.51 years], late-pre-PHV [-1.99 to -1.00 years], circa-PHV [-0.50 to 0.50 years], and post-PHV [> 1.00 years]). All athletes performed a DJ from a 30-cm box onto force plates with performance variables (jump height, ground contact time, and reactive strength index) and absolute and relative kinetic variables during the braking and propulsive phases assessed. Subjects were categorized into GOOD (no impact-peak and spring-like), MODERATE (impact-peak and spring-like), or POOR (impact-peak and not spring-like) stretch-shortening cycle (SSC) function. The post-PHV group exhibited significantly greater values for most absolute kinetic variables compared with early-pre-PHV, late-pre-PHV, and circa-PHV ($p < 0.05$). The differences observed between consecutive maturity groups were similar in magnitude for most absolute variables (Cohen's $d = 0.53$ – 1.70). Post-PHV male athletes outperform their less mature counterparts during a DJ, and this may be attributed to the growth and maturity-related structural and motor control strategy changes that occur in children. Stretch-shortening cycle function in boys seems to improve with maturity status reflected by a greater number of post-PHV participants displaying GOOD SSC function (65.8%) and a greater number of early-pre-PHV participants displaying POOR SSC function (54.4%). However, a number of mature boys displayed POOR SSC function (17.8%), suggesting that the development of SSC function is not exclusively related to maturation.

Key Words: maturation, rebound, stretch-shortening cycle, force-time, boys

Introduction

Muscle function is quite often considered in isometric, concentric, and eccentric actions (37). However, most naturally occurring movements such as running and jumping rely on the stretch-shortening cycle (SSC) (25). This implies that the SSC underpins the performance of a large number of athletic movements, thereby making it an important aspect of athletic development throughout growth and maturation.

Research that has studied the development of SSC function throughout childhood and adolescence has predominantly focused on countermovement jumps (CMJs) and squat jumps (SJs). A recent study by Radnor et al. (46) reported similar CMJ jump heights for pre-PHV and circa-PHV boys, with post-PHV boys exhibiting greater jump heights than their less mature counterparts. This would suggest that jump height tends to increase after the adolescent growth spurt. However, it has been suggested that the CMJ may have a longer amortization phase than other rebound-based activities (26) resulting in the stretch reflex not being activated, and with the SJ lacking an eccentric phase, they do not appropriately represent SSC function (29). To ensure that SSC function is being appropriately assessed, researchers have used rebound-based exercises to elicit preactivation and a stretch reflex in the lower limb (16,25). Among these rebound tasks, the

drop jump (DJ) is one in which the sequence of muscle actions involved and the stretch reflex elicited makes it suitable for the evaluation of SSC function (7). The DJ involves an SSC action with an emphasis on eccentric overload during a phase of triple flexion of the lower limb on landing, followed by subsequent triple extension to launch the body into the air (42), and has been used in previous research to examine SSC function in youth (16,43).

During rebound protocols, jump height, ground contact time, and reactive strength index (RSI), calculated as the ratio of jump height to ground contact time (31), are commonly reported measures to evaluate SSC function (2). However, relying solely on these simple field-based metrics may not reflect the kinetic qualities underpinning SSC function (17) because different jump strategies can affect outcome measures such as jump height (23). Researchers have demonstrated that different strategies can be used in the braking phase, such as greater braking displacement or higher braking peak velocity, to influence the resultant jump performance (34). Given the influence that the braking phase of the SSC has on the propulsive phase and the subsequent outcome (25), it may be necessary to examine multiple braking and propulsive kinetic measures such as to truly understand the determinants of SSC function (33). A more thorough understanding of how kinetic measures influence subsequent performance can help better inform future training prescription (23).

Previous research has examined the effect of increased drop height on subsequent jump performance in youth, and although

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their findings indicated that older youth performed better than their younger counterparts when dropping from greater heights, the study failed to explain how maturity status affects the kinetic variables that drive this performance (16). Although Pedley et al. (43) proposed a novel method to categorize youth into poor, moderate, or good categories based on SSC function, the variables investigated by the authors were limited to mean and peak forces, center of mass displacement, and power in the concentric phase and failed to examine important measures such as impulse, which has been suggested to significantly influence jump performance (24). When examining the effect of maturation on drop jump force-time profiles in girls, Pedley et al. (41) reported an increase in absolute forces throughout maturation, with no significant differences being observed between maturity groups for jump height, ground contact time, and RSI. Additionally, the authors reported that there were no significant differences in the categorization based on SSC function between prepubertal and pubertal participants compared with the whole group, with most girls exhibiting poor SSC function regardless of maturity status (41). It is currently unclear if boys of different maturity status would exhibit similar maturity-related differences in kinetics and in the development of SSC function. Given the sequence of different muscle actions involved during the SSC (50), the changes in motor co-ordination associated with maturation (14), and the difference between boys and girls in maturity-related neuromuscular development (13), it is also important to have a granular understanding of the development of SSC function in boys. Such an understanding of the maturity-related development of the kinetic qualities underpinning SSC performance will help to better inform the design of training programs to improve performance in youth. Therefore, the aim of the current study is to examine the influence of maturity status on DJ kinetics and SSC categorization in young boys. It was hypothesized that the more mature boys would exhibit better DJ kinetics and SSC function compared with their less mature counterparts.

Methods

Experimental Approach to the Problem

This study used a cross-sectional design to assess the differences in DJ performance between early-pre-PHV, late-pre-PHV, circa-PHV, and post-PHV boys. All participants were requested to attend a single testing session, where they performed a DJ from a height of 30 cm onto force plates. A 1-way analysis of variance (ANOVA) was used to identify differences in force-time characteristics during the braking and propulsive phases between maturity groups.

Subjects

Three-hundred forty-one young male athletes (academy soccer and cricket players) volunteered to participate in the study. Written parental consent and subject assent were obtained, and physical readiness questionnaires were completed before commencement of any data collection. Subjects reported no injuries at the familiarization or testing sessions. All subjects engaged in sports practice twice per week and one match weekly, in addition to participation in the schools' physical education program for 2 hours per week. Additionally, although the subjects might have played or trained other sports, this was not recorded. Although the subjects had undergone a 4-week program focussed on developing movement competency, they had not been exposed to a

structured resistance training program at any point before the data being collected. The data for the current study were collected between 2017 and 2019, before the coronavirus pandemic. Ethical approval was granted by the Cardiff Metropolitan University Research Ethics Committee.

Procedures

Anthropometrics. Body mass was recorded to the nearest 0.1 kg on a weighing scale (SECA, 321; Vogel & Halke, Hamburg, Germany). Standing height and sitting height were collected to the nearest 0.1 cm using a stadiometer (SECA, 321; Vogel & Halke, Hamburg, Germany). Maturity offset (years from peak height velocity [PHV]) was calculated by entering anthropometric data into a sex-specific regression equation (Equation 1)

$$\begin{aligned} \text{Maturity Offset } (\pm 0.59 \text{ years}) = & -9.236 + \\ & [0.0002708 \times \text{Leg Length and Sitting Height Interaction}] - \\ & [0.001663 \times \text{Age and Leg Length Interaction}] + \\ & [0.007216 \times \text{Age and Sitting Height Interaction}] + \\ & [0.02292 \times \text{Weight by Height Ratio}] \end{aligned} \quad (1)$$

A maturity offset value of < -2.51 years was classified as early-pre-PHV, -1.99 to -1.00 years indicated that the individual was late-pre-PHV, -0.50 to 0.50 indicated that the individual was circa-PHV, and a score of > 1.00 years indicated that the individual was post-PHV (35). A 6-month boundary between maturity groups was chosen to prevent an individual being incorrectly classified because of the error associated with the prediction model (35). Eighty-two participants were categorized between maturity groups and therefore excluded from further analysis, leaving 259 participants for the final analysis ($n = 57$ early-pre, $n = 80$ late-pre, $n = 49$ circa, and $n = 73$ post). Descriptive statistics for the various maturity groups can be found in Table 1.

Drop Jump Assessment. Subjects completed a 10-minute progressive warm-up, inclusive of light mobilization and activation exercises targeting the main muscle groups of the upper and lower extremities, followed by increasing intensity running drills. During the first session, subjects were familiarized with the DJ movement. Specifically, feedback was provided to subjects during this familiarization session, and subjects were allowed to practice the movement until they demonstrated consistent technical competency over multiple trials, as established by the lead researcher. Formal testing was conducted during the second session. Subjects positioned themselves on top of a 30-cm box and were instructed to stand with toes on the edge, with feet hip width apart and with their hands on their hips (42). On the lead researcher's command, the subjects were instructed to step out from the box onto 2 force plates (Pasco, Roseville, CA), which were positioned 12 cm in front of the box and separated by 8 cm, housed within a custom-built frame. Each foot was required to land on an individual force plate. Upon landing, subjects were instructed to "jump as high and as fast as possible" and then to "stick" the second landing on the force plates. Hands were required to remain on the hips through the entire movement to avoid upper-body interference (28), with legs remaining straight while airborne (30). Failure to fulfill all these requirements resulted in that trial being discounted and another attempt being performed. Vertical ground reaction force data were collected at a rate of 1,000 Hz. Three maximal effort trials were completed, separated by a rest period of 1 minute between each trial. Raw ground reaction force data were smoothed using a fourth-order

Table 1**Mean ($\pm$ SD) values for descriptive details of each maturity group's anthropometric data.***

Maturity group	<i>n</i>	Standing height (cm)	Mass (kg)	Age (y)	Maturity offset (y from PHV)
Early-pre-PHV	57	139.91 $\pm$ 4.75†	34.2 $\pm$ 3.7†	10.54 $\pm$ 0.72†	-2.99 $\pm$ 0.29†
Late-pre-PHV	80	153.29 $\pm$ 6.05†	43.9 $\pm$ 6.9†	12.39 $\pm$ 0.93†	-1.55 $\pm$ 0.28†
Circa-PHV	49	165.23 $\pm$ 6.01†	54.7 $\pm$ 7.4†	14.16 $\pm$ 0.90†	-0.05 $\pm$ 0.31†
Post-PHV	73	177.26 $\pm$ 5.13†	67.2 $\pm$ 7.1†	15.87 $\pm$ 0.86†	1.84 $\pm$ 0.65†

*PHV = peak height velocity.

†Significant between-group difference with all other maturity groups ($p < 0.05$).

recursive low-pass Butterworth filter with a cutoff of 30 Hz and using a bespoke Microsoft Excel Spreadsheet (Microsoft Corporation, 2016). The variables were automatically calculated from the force data of the first ground contact period and grouped into subcategories depending on the qualities they represented: performance variables, peak force variables, power variables, impulse variables, and movement strategy variables (Table 2).

The absolute variables were allometrically scaled to obtain relative values (Table 2). Allometric scaling provides a normalized methodological approach for performance tests (22) and has previously been used to scale measurements of full-body strength in youth of difference body size (9). Allometric normalization involves dividing the absolute force from a performance test by body mass in kilograms^{0.67} (11). However, given the limitations of allometrically scaling separate groups using a fitted exponent (51), all relative variables in the current study were calculated by allometric scaling using a population-specific exponent.

Subjects were categorized based on presentation of an impact peak in their force-time profile (the highest transient, visible force peak during the braking phase of ground contact and occurring within the first 20% of ground contact) (20,38), and whether they display spring-like behavior (Pearson's product-moment correlation between vertical ground reaction force and vertical center of mass displacement through the entire contact phase being < -0.80) (40,43). As in the study by Pedley et al. (43), participants were then classified into GOOD (no impact peak and still spring-like), MODERATE (impact peak and still spring-like), or POOR (impact peak and not spring-like). Individual force traces were time-normalized to 101 data points using a custom Microsoft Excel spreadsheet. Mean $\pm$ SDs of the vertical forces were calculated for each time point to derive average force traces for each SSC category.

Statistical Analyses

Between-Group Comparisons. Data were assessed for normality using the Shapiro-Wilk test. For normally distributed data, a 1-way ANOVA was used to determine differences for all DJ kinetic variables across maturity groups. Post hoc analysis was used to identify any significantly between-maturity group differences using either Bonferroni or Games-Howell tests, where equal variances were or were not assumed, respectively. A Kruskal-Wallis test was performed to identify differences between maturity groups when variables were not normally distributed. Dunn's pairwise test adjustment with the Bonferroni correction were applied post hoc to determine where significant differences occurred. The level of significance was set at alpha level $p < 0.05$. Cohen's d effect sizes were calculated to determine the magnitude of between-group differences that were observed (10) and were categorized as follows: < 0.19 = trivial, 0.20 – 0.59 = small, 0.60 – 1.19 = moderate, 1.20 – 1.99 = large, 2.00 – 3.99 = very large, and > 4.00 = extremely large (19).

Chi-Square Analysis. Chi-square (χ^2) analysis was used to investigate between-group differences for the number of individuals within each SSC function category (POOR, MODERATE, and GOOD). In the Chi-square test, analysis of the adjusted standardized residuals was completed to identify frequencies that were > 1.96 z -scores ($p < 0.05$) different to the whole group distribution. Adjusted residuals were converted into Chi-square values and subsequently into p values. The Bonferroni correction was used to produce an adjusted alpha level of $p < 0.05$ to reduce the potential for a type I error as a result of multiple comparisons (4).

All ANOVAs, Chi-square analyses, and post hoc analyses were processed using SPSS (v24.0.0.2.; Chicago, Illinois), whereas descriptive statistics and effect sizes were computed using Microsoft Excel. ICC, CV%, and typical error were calculated using an online spreadsheet run through Microsoft Excel (18).

Results

Between-Group Comparisons

Drop Jump Performance Variables. The performance variables for the 4 maturity groups are shown in Figure 1. No differences were observed from early-pre-PHV to late-pre-PHV and from late-pre-PHV to circa-PHV ($p > 0.05$) for jump height; however, the post-PHV group displayed significantly greater jump heights than early-pre-PHV, late-pre-PHV, and circa-PHV groups ($d = 1.43, 1.12$, and 0.55 , respectively; $p < 0.05$). The circa-PHV exhibited greater jump height than early-pre-PHV ($d = 0.77$; $p < 0.05$). The late-pre-PHV, circa-PHV, and post-PHV groups had a greater RSI than the early-pre-PHV group ($d = 0.53, 0.92$, and 1.08 , respectively; $p < 0.05$), whereas the post-PHV cohort also had a greater RSI than the late-pre-PHV group ($d = 0.51$; $p < 0.05$). The changes observed from early-pre-PHV to late-pre-PHV, late-pre-PHV to circa-PHV, and circa-PHV to post-PHV for jump height and RSI seem to be similar in magnitude (Figure 1). No significant differences were observed between late-pre-PHV and circa-PHV and between circa-PHV and post-PHV for RSI ($p > 0.05$). Finally, there were no significant differences observed between any groups for ground contact time ($p > 0.05$).

Force Variables. Table 3 shows the force variables for the 4 maturity groups. Late-pre-PHV, circa-PHV, and post-PHV cohorts had significantly greater mean braking force ($d = 1.21, 2.29$, and 2.92 , respectively; $p < 0.05$), peak braking force ($d = 0.99, 1.43$, and 1.94 , respectively; $p < 0.05$), mean propulsive force ($d = 1.58, 3.21$, and 3.92 , respectively; $p < 0.05$), and propulsive peak force ($d = 1.16, 2.38$, and 2.75 , respectively; $p < 0.05$) compared with the early-pre-PHV cohort. The circa-PHV and post-PHV groups exhibited a significantly greater mean braking force ($d = 0.83$ and 1.47 , respectively; $p < 0.05$), mean propulsive force ($d = 1.29$ and 2.33 , respectively; $p < 0.05$), and propulsive peak force ($d = 0.84$ and 1.39 , respectively; $p < 0.05$) than late-pre-PHV.

Table 2**List of variables investigated and the method of calculation.**

Variable	Method of Calculation
Performance variables	
Ground contact time (s)	The duration of ground contact time of the first landing was established by determining the time interval between the first (initial contact) and final (take-off) data point using the threshold of $5 \times \text{SD}$ of noise.
Jump height (m)	Defined as the largest vertical displacement of the body's center of mass during the flight time between the first and the second ground contact, calculated using jump height = $\text{TOV}^2/2g$ (TOV = vertical velocity of the center of mass at take-off; g = acceleration because of gravity).
Reactive strength index	Calculated as the ratio of jump height (mm) and first ground contact time (ms).
Peak force variables	
Mean peak braking force (N)	The ground reaction force at each timing sample was summed, and the mean braking force was calculated as the average force between initial ground contact and the timing of the maximal displacement of center of mass.
peak braking force (N)	Calculated as the highest force value, which was followed by a decline in force, during the braking phase of the first ground contact period. The braking phase was identified as the phase that followed the timing of initial contact up until the peak center of mass displacement.
Mean propulsive force (N)	The ground reaction force at each timing sample was summed, and the mean propulsive force was calculated as the average force from the timing of lowest point of the center of mass and the point of take-off.
Propulsive peak force (N)	Calculated as the highest force value recorded during the propulsive phase of the movement. The propulsive phase was identified as the phase that followed the timing of the lowest point of the center of mass up until take-off.
Impulse variables	
Net impulse (N-s)	Net impulse was measured by subtracting body mass impulse from total vertical impulse.
Braking impulse (Ns)	Braking impulse is the total vertical impulse applied to the ground during the landing phase of ground contact and is calculated by summing the instantaneous impulse values between initial ground contact and peak center of mass displacement.
Propulsive impulse (Ns)	Propulsive impulse is calculated by summing the instantaneous impulse values between peak center of mass displacement and take-off phase of ground contact.
Power variables	
Mean braking power (W)	Velocity of the center of mass was multiplied by the ground reaction force at each timing sample to derive power. Mean braking power was calculated as the average power between initial ground contact and the timing of the maximal displacement of center of mass.
Mean propulsive power (W)	Velocity of the center of mass was multiplied by the ground reaction force at each timing sample to derive power. Mean propulsive power was calculated as the average power from the timing of lowest point of the center of mass and the point of take-off.
Movement strategy variables	
Spring-like correlation	Determined by calculating a Pearson's product-moment correlation coefficient using center of mass displacement and absolute vertical force throughout the initial ground contact period.
Peak center of mass displacement (m)	Determined by dividing the vertical component of the ground reaction force by the participant's body mass to determine acceleration. Double integration of acceleration was then performed to calculate displacement of center of mass and the peak was then obtained by identifying the minimum value.
Braking mean center of mass velocity (m/s)	Determined by dividing the vertical component of the ground reaction force during the braking phase by the participant's body mass to determine acceleration. Integration of acceleration was then performed to calculate velocity of COM during the braking phase.
Propulsive mean center of mass velocity (m/s)	Determined by dividing the vertical component of the ground reaction force during the propulsive phase by the participant's body mass to determine acceleration. Integration of acceleration was then performed to calculate velocity of COM during the propulsive phase.
Relative variables	
Relative peak braking force ($\text{N/kg}^{0.74}$)	The absolute variables were normalized using allometric scaling (P/BM^b , where P represents the variable, BM is body mass in kilograms, and b is the power exponent). The power exponent for allometric scaling was derived from the data collected, by plotting log-transformed data of each variable and body mass on a scale and then using the slope of the linear regression line.
Relative propulsive peak force ($\text{N/kg}^{1.09}$)	
Relative net impulse ($\text{N.s/kg}^{1.03}$)	
Relative braking impulse ($\text{N.s/kg}^{1.05}$)	
Relative propulsive impulse ($\text{N.s/kg}^{0.96}$)	
Relative mean braking power ($\text{W/kg}^{1.04}$)	
Relative mean propulsive power ($\text{W/kg}^{1.12}$)	

However, only the post-PHV group displayed a significantly greater peak braking force than late-pre-PHV ($d = 0.77$; $p < 0.05$). The post-PHV cohort also had significantly greater mean braking force, mean propulsive force, and propulsive peak force compared with circa-PHV ($d = 0.72, 1.28$, and 0.68 , respectively; $p < 0.05$). The changes observed from early-pre-PHV to late-pre-PHV for peak braking force and peak propulsive force seem to be larger in magnitude compared with the changes from late-pre-

PHV to circa-PHV and circa-PHV to post-PHV (Table 3). The late-pre-PHV group presented significantly greater relative peak braking force compared with post-PHV ($d = 0.49$; $p < 0.05$; Table 3). No significant differences were observed between maturity groups for relative propulsive peak force ($p > 0.05$).

Impulse Variables. The effect of maturity status on impulse variables is shown in Table 3. The late-pre-PHV, circa-PHV, and

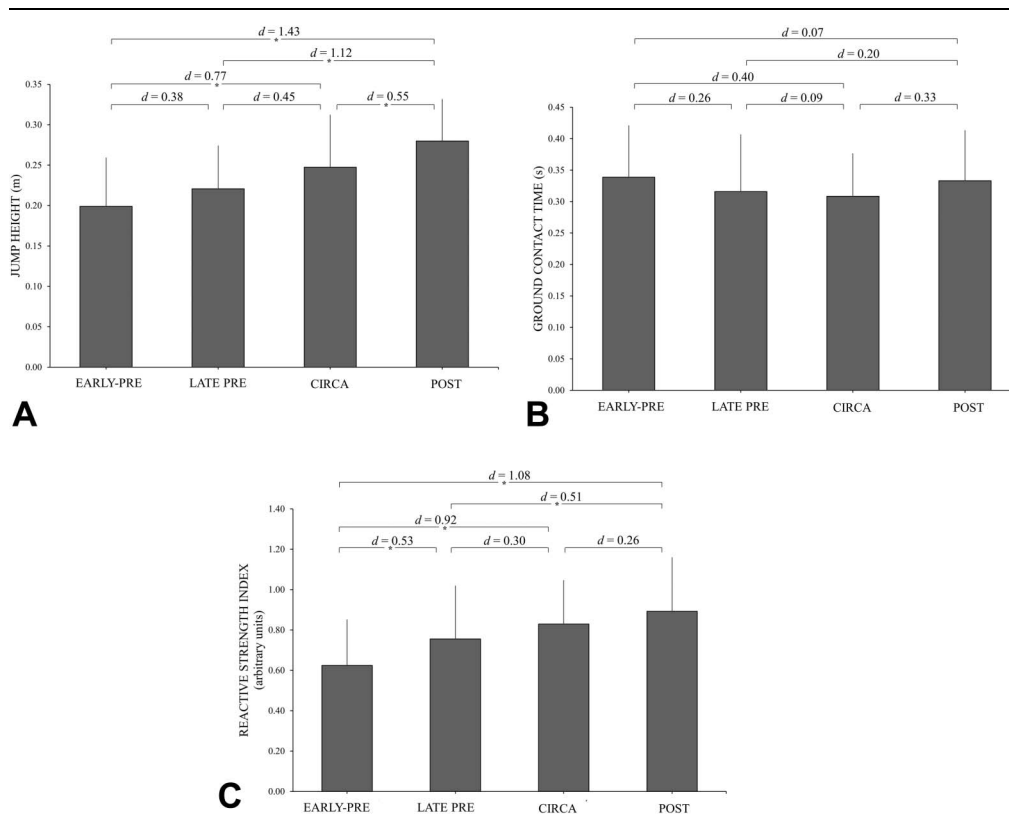


Figure 1. Effect of maturity status on (A) jump height, (B) ground contact time, and (C) RSI. d = between group effect size. *Significant difference between maturity groups ($p < 0.05$).

post-PHV groups displayed significantly greater net impulse ($d = 1.47, 2.94$, and 4.18 , respectively; $p < 0.05$), braking impulse ($d = 1.17, 2.56$, and 3.93 , respectively; $p < 0.05$), and propulsive impulse ($d = 1.21, 2.64$, and 3.93 , respectively; $p < 0.05$) compared with the early-pre-PHV group. The post-PHV group exhibited significantly greater net impulse ($d = 2.95$ and 1.66 ; $p < 0.05$), braking impulse ($d = 2.80$ and 1.70 ; $p < 0.05$), and propulsive impulse ($d = 2.64$ and 1.68 ; $p < 0.05$) compared with the late-pre-PHV and circa-PHV cohorts. The circa-PHV group displayed significantly greater net, braking, and propulsive impulse compared with the late-pre-PHV group ($d = 1.49, 1.28$, and 1.22 , respectively; $p < 0.05$). The changes observed in all the absolute impulse variables from early-pre-PHV to late-pre-PHV, late-pre-PHV to circa-PHV, and circa-PHV to post-PHV seem to be similar in magnitude (Table 3). No significant differences were observed between groups for any of the relative impulse variables ($p > 0.05$; Table 3).

Power Variables. Table 3 shows the power variables for the 4 maturity groups. Analysis revealed that the late-pre-PHV, circa-PHV, and post-PHV groups displayed significantly greater absolute mean braking power ($d = 1.05, 2.10$, and 2.58 , respectively; $p < 0.05$) and mean propulsive power ($d = 1.18, 2.40$, and 3.11 , respectively; $p < 0.05$) than the early-pre-PHV group. The post-PHV group presented significantly greater absolute mean braking power ($d = 1.49$ and 0.82 , respectively; $p < 0.05$) and mean propulsive power ($d = 1.97$ and 1.10 , respectively; $p < 0.05$) than the late-pre-PHV and circa-PHV groups. The circa-PHV cohort exhibited significantly greater absolute mean braking and propulsive power than the late-pre-PHV group ($d = 0.84$ and 1.05 , respectively; $p < 0.05$). The changes observed in the

absolute power variables from early-pre-PHV to late-pre-PHV, late-pre-PHV to circa-PHV, and circa-PHV to post-PHV seem to be similar in magnitude (Table 3). No significant differences across maturity groups were observed for relative mean braking power and relative mean propulsive power ($p > 0.05$; Table 3).

Landing Strategy Variables. The landing strategy variables for the maturity groups are shown in Table 3. Analysis revealed that the late-pre-PHV, circa-PHV, and post-PHV groups displayed significantly greater spring-like correlation ($d = 0.51, 1.18$, and 0.98 , respectively; $p < 0.05$) compared with the early-pre-PHV group. The post-PHV group exhibited a greater peak COM displacement than the early-pre-PHV, late-pre-PHV, and circa-PHV groups ($d = 0.90, 0.86$, and 0.49 , respectively; $p < 0.05$) and significantly greater spring-like correlation than the late-pre-PHV group ($d = 0.43$; $p < 0.05$). The only significant difference between late-pre-PHV and circa-PHV groups was for spring-like correlation ($d = 0.55$; $p < 0.05$), with the circa-PHV group exhibiting a greater value. Although the circa-PHV and post-PHV groups displayed a significantly greater eccentric COM velocity ($d = 0.54$ and 0.80 , respectively; $p < 0.05$) compared with the early-pre-PHV group, only the post-PHV group had a greater eccentric COM velocity than late-pre-PHV group ($d = 0.51$; $p < 0.05$). No other differences were observed between groups ($p > 0.05$) for the movement strategy variables.

Stretch-Shortening Cycle Categories

When analyzing the whole group, 49.0% ($n = 127$) of the participants were categorized as having GOOD SSC function, 20.5% ($n = 53$) as MODERATE function, and 30.5% ($n = 79$) as POOR

Table 3**Effect of maturity status on absolute and relative force, impulse, power, and movement strategy variables.***

Variable	Maturity group				Between-group effect sizes (d)					
	Early-pre (I)	Late-pre (II)	Circa (III)	Post (IV)	I-II	I-III	I-IV	II-III	II-IV	III-IV
Force variables										
Mean braking force (N)	850 ± 194	1,166 ± 312†	1,413 ± 288†‡	1,633 ± 326§	1.21	2.29	2.92	0.83	1.47	0.72
Peak braking force (N)	1,490 ± 408	1988 ± 583†	2,219 ± 599†	2,425 ± 548†‡	0.99	1.43	1.94	0.39	0.77	0.36
Relative braking peak (N/kg ^{0.74})	108.3 ± 26.6	120.8 ± 30.6	114.4 ± 27.9	107.4 ± 23.3‡	0.44	0.23	0.04	0.22	0.49	0.27
Mean propulsive force (N)	730 ± 123	1,006 ± 214†	1,277 ± 206†‡	1,595 ± 286§	1.58	3.21	3.92	1.29	2.33	1.28
Propulsive peak force (N)	1,033 ± 254	1,499 ± 510†	1904 ± 452†‡	2,258 ± 577§	1.16	2.38	2.75	0.84	1.39	0.68
Relative propulsive peak (N/kg ^{1.09})	21.8 ± 5.3	24.1 ± 7.0	24.0 ± 5.0	22.6 ± 5.3	0.38	0.43	0.17	0.01	0.23	0.26
Force at peak COMD	978 ± 260	1,421 ± 524†	1825 ± 468†‡	2,162 ± 621§	1.07	2.24	2.49	0.81	1.29	0.61
Relative force at peak COMD (N/kg ^{1.11})	19.3 ± 5.1	21.2 ± 6.8	21.3 ± 4.9	20.0 ± 5.3	0.33	0.42	0.14	0.02	0.20	0.26
Impulse variables										
Net impulse (N.s)	140 ± 30	186 ± 31†	233 ± 32†‡	296 ± 43§	1.47	2.94	4.18	1.49	2.95	1.66
Relative net impulse (N.s/kg ^{1.03})	3.66 ± 0.70	3.78 ± 0.52	3.78 ± 0.54	3.86 ± 0.51	0.20	0.20	0.34	0.01	0.16	0.15
Braking impulse (N.s)	121 ± 24	154 ± 30†	194 ± 31†‡	258 ± 42§	1.17	2.56	3.93	1.28	2.80	1.70
Relative braking impulse (N.s/kg ^{1.05})	2.96 ± 0.50	2.90 ± 0.43	2.91 ± 0.50	3.09 ± 0.47	0.13	0.10	0.28	0.02	0.43	0.37
Propulsive impulse (N.s)	134 ± 27	169 ± 32†	205 ± 27†‡	259 ± 37§	1.21	2.64	3.93	1.22	2.64	1.68
Relative propulsive impulse (N.s/kg ^{0.96})	4.46 ± 0.66	4.48 ± 0.60	4.41 ± 0.54	4.56 ± 0.58	0.03	0.09	0.16	0.12	0.14	0.27
Power variables										
Mean braking power (W)	943 ± 335	1,341 ± 415†	1,664 ± 351†‡	2009 ± 477§	1.05	2.10	2.58	0.84	1.49	0.82
Relative braking power (W/kg ^{1.04})	23.8 ± 8.4	26.3 ± 7.4	26.0 ± 5.6	25.1 ± 6.2†	0.32	0.31	0.18	0.04	0.17	0.14
Mean propulsive power (W)	889 ± 280	1,259 ± 343†	1,609 ± 320†‡	2032 ± 437§	1.18	2.40	3.11	1.05	1.97	1.10
Relative propulsive power (W/kg ^{1.21})	16.9 ± 5.2	18.3 ± 4.9	18.3 ± 3.8	18.2 ± 3.9	0.28	0.30	0.27	0.01	0.03	0.03
Movement strategy variables										
Spring-like correlation	0.74 ± 0.15	0.81 ± 0.15†	0.88 ± 0.09†‡	0.87 ± 0.13†‡	0.51	1.18	0.98	0.55	0.43	0.08
Peak COMD (m)	0.18 ± 0.6	0.18 ± 0.06	0.20 ± 0.07	0.24 ± 0.07§	0.06	0.35	0.90	0.30	0.86	0.49
Mean braking COM velocity (m/s)	1.09 ± 0.20	1.15 ± 0.16	1.19 ± 0.16†	1.23 ± 0.15†‡	0.33	0.54	0.80	0.24	0.51	0.26
Mean propulsive COM velocity (m/s)	1.20 ± 0.18	1.24 ± 0.15	1.26 ± 0.14	1.27 ± 0.14	0.26	0.38	0.43	0.13	0.20	0.07

*COM = center of mass; COMD = center of mass displacement; RFD = rate of force development.

†Significant between-group difference with early-pre-PHV ($p < 0.05$).‡Significant between-group difference with pre-PHV ($p < 0.05$).§Significant between-group difference with all other maturity groups ($p < 0.05$).

function (Figure 2). This is also reflected in the distribution of SSC categories, with analysis revealing a significant interaction between SSC function and maturity status, with more early-pre-PHV group displaying POOR SSC function ($p < 0.05$) and proportionately less early-pre-PHV group exhibiting GOOD SSC function ($p < 0.05$) when compared with the whole group. The opposite was observed for the post-PHV group with proportionately lower number of participants displaying POOR SSC function ($p < 0.05$) and a greater number of participants presenting GOOD SSC function ($p < 0.05$) than the whole group. Although the circa-PHV group had lower number of participants displaying POOR SSC function in comparison with the entire sample ($p < 0.05$), they showed no significantly different distribution among the other 2 SSC categories compared with the whole group ($p > 0.05$). The late-pre-PHV group showed no significantly different distribution among any of the SSC categories when compared with the entire sample ($p > 0.05$).

Discussion

The aim of the current study was to examine the effect of maturity status on DJ kinetics in young boys, by assessing kinetic variables during the braking and propulsive phases of a DJ. The main finding of the current study was that the post-PHV group exhibited greater values for all absolute kinetic variables, other than peak braking force and COM velocities, compared with early-pre-PHV, late-pre-PHV, and circa-PHV. The differences

observed between early-pre-PHV to late-pre-PHV, late-pre-PHV to circa-PHV, and circa-PHV to post-PHV were similar in magnitude for most variables. SSC function seems to improve with maturity status, as demonstrated by a greater number of post-PHV participants displaying GOOD SSC function and greater number of early-pre-PHV participants displaying POOR SSC function, when compared with the whole group.

Unlike the findings of Pedley et al. (41), the current study observed greater jump height and RSI, with no change in ground contact time, in the more mature participants compared with the less mature cohorts. The findings of the current study are in conjunction with previous research reporting that performance during vertical during and rebound activities improves as boys mature (44), but with limited changes in ground contact time (5). The gender-based difference in maturity-related development of vertical jump performance can be attributed to the divergence in development of neuromuscular patterns during maturation (44). The performance in the propulsive phase of the DJ has been suggested to be strongly influenced by the force producing capabilities of the muscle (7). Based on this, the greater jump height and RSI in the more mature boys compared with their less mature counterparts can be explained by their greater absolute propulsive mean and peak force, net impulse, propulsive impulse, and propulsive power. Such improvements in force producing capabilities with maturity have been associated with increases in muscle mass (46), with maturity-related increases in vastus lateralis muscle thickness being suggested to underpin vertical jump

performance (46). This suggested influence of muscle mass on force-producing capabilities is further reinforced by the lack of differences between the maturity groups for propulsive peak force relative to body mass, which is in congruence with the findings of Radnor et al. (46).

The more mature participants exhibited significantly greater absolute mean and peak braking forces compared with other cohorts, with the post-PHV group displaying significantly greater peak braking force than late-pre-PHV group, even when normalized to body mass. The large amounts of braking loading during DJs (8) have been suggest to result in high stretch loads (6,26) and result in long ground contact times (1). Although increased braking loading has been linked to improved jump performance in adults (6,7), it has been suggested that youth may not respond the same way because of their underdeveloped skeletal musculature and the preloading potentially exceeding the SSC capabilities of their lower limb musculature (16). Although the ground contact times remained unchanged between maturity groups, the more mature boys seem to accommodate their greater absolute peak braking force and impulse by displacing more during ground contact, with a greater braking COM velocity, thereby resulting in greater mean braking power. Although this is indicative of an increased and faster prestretch, suggestive of storage of a greater amount of elastic energy (15), the greater displacement also allows for a greater distance over which force can be developed (21). These mechanisms are reflected by the maturity status-related increases in absolute mean and peak forces, power, and more importantly net impulse, which is indicative of change in momentum (48), and therefore increased jump height (49).

The findings of the current study are consistent with previous research showing an improvement in SSC function associated

with maturity status (43,44), reflected by a greater proportion of boys in the post-PHV group having GOOD SSC function and greater proportion in the pre-PHV group having POOR SSC function, when compared with the whole group. However, the findings are unlike those reported in girls by Pedley et al. (41), with the differences between boys and girls potentially being because of gender-related differences in the development of lower extremity energy absorption, stiffness, and muscle strength with maturation (44). Based on the study by Padua et al. (40), who suggested a cutoff of >0.80 to describe whether an individual is displaying spring-like behavior, the higher spring-like correlation observed in the more mature participants in the current study indicates that spring-like behavior might improve with maturation. Although the greater spring-like correlation in the more mature participants might be indicative of greater levels of stiffness, they also displayed greater levels of peak COM displacement than their cohorts. With the absolute mean braking forces being greater in the more mature participants, their greater COM displacement might be a movement strategy to attenuate the ground reaction forces, with it being suggested that greater flexion of the ankles, knees, and hips allows for more time to distribute impact forces and allow for the absorption of these forces by the musculature (32). Neural mechanisms such as pre-activation and amplitude and timing of stretch-reflex have also been suggested to influence lower limb performance during rebound activities (39). Additionally, lower levels of preactivation have been suggested to result in peak vertical ground reaction forces that manifest as impact peaks during the early phase of ground contact (43). Although the distribution within the SSC categories seems to become stable between pre-PHV and circa-PHV, potentially be explained by the maturity-related improvements in neural mechanisms, this is something that warrants

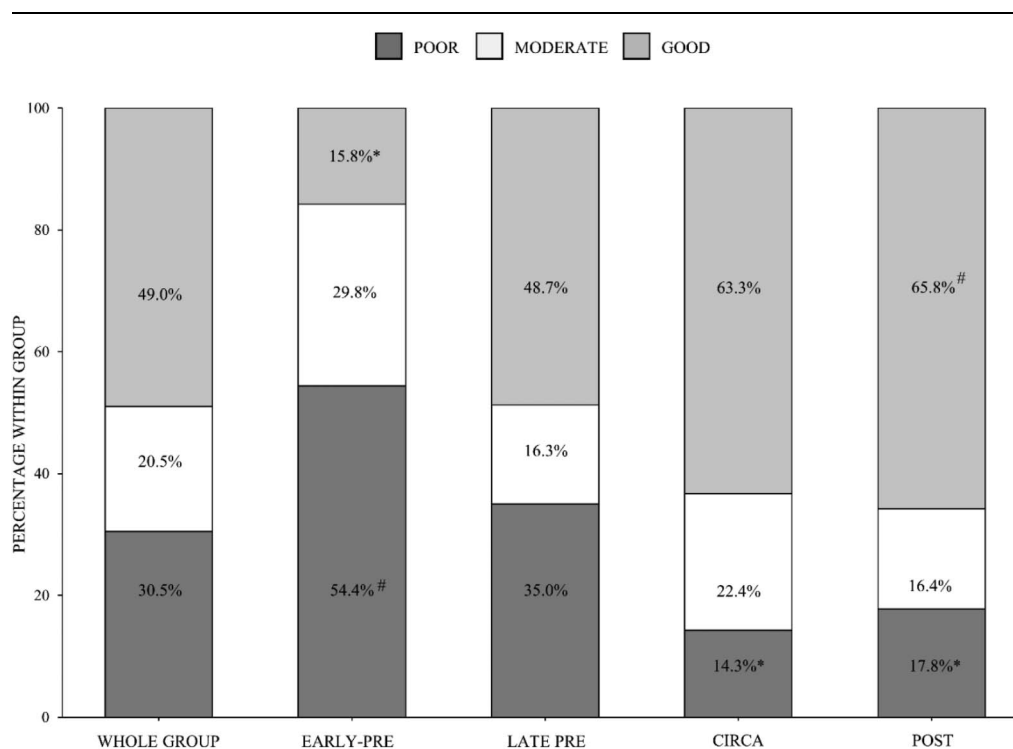


Figure 2. Proportion of stretch-shortening cycle function categories within and between different stages of maturation (early-pre-PHV, $n = 57$; late-pre-PHV, $n = 80$; circa-PHV, $n = 49$; post-PHV, $n = 73$). [#]Significantly more in proportion to the whole group ($p < 0.05$). ^{*}Significantly less in proportion to the whole group ($p < 0.05$).

further examination. It is important to note that, similar to the findings of Pedley et al. (43), although several pre-PHV boys displayed GOOD SSC function, a small proportion of the post-PHV group displayed POOR SSC function, suggesting that SSC development was not exclusively a function of maturation.

Children experience various maturity-related changes including increases in muscle volume and thickness, fascicle length, tendon cross-sectional area and stiffness, and improvements in muscle activation, preactivation, and stretch reflex control (52). These changes drive increases in strength and power (36), subsequently influencing sprint (36) and jump (45) performance, which are underpinned by the stretch-shortening cycle (12). However, it is also important to acknowledge that the nature of the sport engaged in can also have an influence on this development. For example, Latorre Román et al. (27), examining the effect of contrast training in youth basketball players, reported a significant increase in jump height in the intervention group and the control group at postintervention testing. The improvements exhibited by the control group could be attributed to their basketball sessions, with data suggesting that during a match, youth basketball players can perform up to 2 jumps per minute (47). This would suggest that between training and matches, participants are exposed to a high amount of neuromuscular stimulus, potentially increasing with age, which could also have played a role in the differences observed in the current study.

It should be acknowledged that optimal drop height varies based on individual capabilities. Previous research has suggested drop heights between 20 and 40 cm for youth (5). Although Bassa et al. (3) did not specify an optimal drop height for youth, they did suggest low drop heights for younger children. Based on this, it could be speculated that the drop height of 30 cm used within the study might have exceeded the capabilities of the pre-PHV boys. Additionally, the authors did not have details of any activities engaged in outside of the study that might have influenced the findings. Future research should aim to document such information.

Practical Applications

The findings of the study reinforce the need for the examination of SSC function to move beyond just measures of jump height and RSI. Although previous research has frequently used RSI and jump height to assess SSC function, the current study showed no significant differences in RSI from late-pre-PHV to circa-PHV and circa-PHV to post-PHV and no difference in jump height from late-pre-PHV to circa-PHV. However, small-to-moderate significant differences were observed in absolute mean and peak forces, power, and impulse from late-pre-PHV to circa-PHV and circa-PHV to post-PHV. Different jump strategies have been suggested to affect outcome measures such as jump height and RSI, and the differences observed in measures such as mean and peak forces, impulse, and power provide a better understanding of the varying kinetic and movement strategies driving performance in the different maturity groups. It is important for practitioners to refrain from relying solely on outcome measures when assessing performance or the efficiency of a training program because they may not reflect the underlying mechanisms. The authors acknowledge that most youth coaches might not have access to force plates, so future research should aim to examine what underlying kinetic qualities are best represented by these outcome measures. For example, jump

height has been shown to be strongly related to net vertical impulse ($r > 0.93$), albeit from countermovement jumps and squat jumps (24). An understanding of the kinetic measures represented would better inform the design of training programs to target the improvement-specific qualities. Although the current study highlights the development of physical qualities with maturation, several mature boys displayed POOR SSC function, suggesting that the development of SSC function is not exclusively related to maturation. Therefore, practitioners should take into account individual capabilities and not just maturity status when prescribing plyometric exercises.

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